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SECTION 33 26 00

RELIEF WELLS

PART 1 GENERAL

1.1 SCOPE

The work provided for herein consists of furnishing all plant, labor, material, and equipment, and performing all work associated with the relief wells, as specified herein, incidental thereto, and in accordance with the drawings. This work includes, drilling the well boreholes, setting casing, assembling each relief well, hanging and centering each relief well in the borehole, and placement of filter pack and sealing and grouting materials - all to the elevations shown on the plans, according to each well detail, and within the tolerances permitted. The work also includes relief well development, pump and sand testing, video inspections, chlorine treatments, fabricating check valves, disposal of materials, piezometer abandonment, protecting the work from damage, performing the work in a manner that prevents damage, and providing all associated quality control, testing, and documentation - including the timely submission of accurate and complete preconstruction submittals, shop drawings, product data, test reports, certificates, and quality control reports. The work also requires the use of temporary risers and temporary casing during relief well construction and managing the work to accommodate the groundwater conditions that vary with location and are dependent on the reservoir stage. This section also covers the work for relief well details that have not yet been provided, as denoted by elevations of materials that are "TBD" on the relief well profile drawings. Refer to Section 01 11 00 SUMMARY OF WORK for details on how the Government will provide this information and incorporate their release into the bid to perform this work.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA A100

(2020) Water Wells

AWWA B100

(2016; Errata 2017; Addenda 2018) Granular Filter Material

AMERICAN WELDING SOCIETY (AWS)

AWS D1.2/D1.2M

(2014; Errata 1 2014; Errata 2 2020)
Structural Welding Code - Aluminum

ASTM INTERNATIONAL (ASTM)

ASTM A53/A53M

(2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated,

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Welded and Seamless

ASTM A276/A276M

(2024) Standard Specification for
Stainless Steel Bars and Shapes

ASTM A312/A312M

(2022b) Standard Specification for
Seamless, Welded, and Heavily Cold Worked
Austenitic Stainless Steel Pipes

ASTM A555/A555M

(2023) Standard Specification for General
Requirements for Stainless Steel Wire and
Wire Rods

ASTM C136/C136M

(2019) Standard Test Method for Sieve
Analysis of Fine and Coarse Aggregates

ASTM D75/D75M

(2019) Standard Practice for Sampling
Aggregates

ASTM D2240

(2015; E 2017; R 2021) Standard Test
Method for Rubber Property - Durometer
Hardness

FLORIDA ADMINISTRATIVE CODE (F.A.C.)

Chapter 62-531 F.A.C.

Water Well Contractor Licensing
Requirements

Chapter 62-532 F.A.C.

Water Well Permitting and Construction

NSF INTERNATIONAL (NSF)

NSF/ANSI 61

(2022) Drinking Water System Components -
Health Effects

1.3 DEFINITIONS

1.3.1 Coefficient of Uniformity (Cu)

Coefficient of Uniformity (Cu) is a ratio calculated from the results of grain size distribution test performed on a representative sample in accordance with ASTM C136/C136M and calculated by determining the grain size equal to 60 percent passing divided by the grain size equal to 10 percent passing ($Cu = D_{60}/D_{10}$). D_{60} and D_{10} values are calculated by performing a log-linear interpolation of the of the grain size distribution test result.

1.3.2 Effective Size

Effective size is the size of the particle equal to the 10 percent passing of a grain size distribution performed on a representative sample in accordance with ASTM C136/C136M. Demonstrated compliance is by scaled graphical representation of the grain size distribution falls within the minimum and maximum limits and by comparison of the specified values to the 10 percent passing particle size calculated from a log-linear interpolation of the particle size equal to 10 percent passing of the grain size distribution test result.

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1.4 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Installation Plan; G, D0

SD-02 Shop Drawings

Relief Well Components and Installation Devices; G, D0

Surge Block; G, D0

Well Installation Diagram; G, D0

SD-03 Product Data

Drilling Fluid Additives; G, D0

Well Screens; G, D0

Grout; G, D0

Permanent Steel Casing; G, D0

SD-06 Test Reports

Filter Pack Gradations; G, D0

Well Development Data; G, D0

Pump Test and Sand Test Results; G, D0

Video Inspection; G, D0

SD-07 Certificates

Well Driller's License; G, D0

Filter Pack Source; G, D0

1.5 QUALITY CONTROL

Perform all inspection, sampling, testing, corrective actions, and reporting required to demonstrate compliance with the technical provisions of this specification. Ensure that the appropriate quality control (QC) personnel are present and active throughout each stage of the construction process. Unless otherwise stated, ensure the elevation of changes between materials are to the nearest 0.1 foot. Provide vertical survey control for temporary construction datums that is accurate to 0.01 foot of elevation for measuring depths to document well construction. Check and ensure that all depth measurements are accurately made from each measurement datum. Report all measurement datum elevations, depth measurements, and their conversion to elevation. Before well construction

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begins, all submittals and materials required for well completion must be on hand at the work site, the well screen and riser checked for proper material, length, and slot openings, and filter material inspected and checked against specifications. Well drilling and development equipment and downhole tools must arrive on site clean and decontaminated.

1.5.1 Inspection

The Contractor's QC organization is responsible for inspecting the work as needed to guarantee compliance with each and every requirement of this specification, including but not limited to: survey layout, procurement of materials, transport and storage of materials, well drilling, well assembly, filter pack placement, well development, pump and sand testing, chemical treatment of wells, well backfill, well head construction, protecting the work, and all work incidental thereto.

1.5.2 Sampling, Testing and Documentation

Sample and test individual components of the work as follows:

- a. Prior to the delivery of any filter pack to the site, provide manufacturer certification and documentation of the proposed **filter pack source** from the manufacturer that demonstrates the filter pack meets the requirements of paragraph FILTER PACK. Also include three filter pack gradations for each type of filter pack in accordance with the requirements for filter pack gradations.
- b. Sample the filter pack according to **ASTM D75/D75M** and **AWWA B100** after delivery to the site and prior to placement. Subject the representative samples of the filter pack to grain size analysis testing in accordance with **ASTM C136/C136M** and provide the **filter pack gradations** for each bulk bag (3,000 or 4,000 lb. bag), or every 3 cubic yards of delivered in other increments. Submit the results and receive approval prior to use. The Contractor is responsible for performing quality checks at the source, prior to delivery. Immediately notify the Contracting Officer if the gradation results do not meet the requirements for the respective filter pack of paragraph FILTER PACK.

(1) Maintain all filter pack gradations in a continuous table in .xlsm or .xlsb format that includes only filter pack gradation testing with all information required by required by **ASTM C136/C136M** as well as other metadata associated with the sample and the materials use on the project. Provide the continuous table in the required format each time the gradation results are submitted.

- c. Conduct the development of each well and provide the well development data, as required by under paragraph DEVELOPMENT.
- d. Conduct the pump testing and sand testing of each well and document the results for each well, as required under paragraph PUMP AND SAND TESTING.
- e. Document the installation of each well and provide a well installation diagram in accordance with paragraph WELL INSTALLATION DIAGRAM. Maintain the well construction diagram as the well is being constructed.
- f. Conduct and document the video inspection of each well in accordance

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with paragraph VIDEO INSPECTON.

The results and documentation of each of these is considered an essential component conducting the work prescribed in this section. Deliver all results in the required format in an organized, complete, accurate, neat and in a timely manner. Keep all test results and well construction documentation up to date with the work and daily quality control reports and make the information available to Government immediately upon request.

1.5.3 Action Required

When quality control or quality assurance monitoring or testing detects non-conformance with specifications, take corrective action immediately. Report the details of the irregularities and the actions taken to correct them immediately to the Contracting Officer and include in the daily Quality Control report. The corrective action must include steps taken to prevent recurrence of the irregularity.

1.5.4 Licenses and Permits

Obtain any local, state, or Federal permits required to drill or install relief wells in the state where the work is performed. Submit documentation Of Well Construction Permits to the Contracting Officer and keep on file at the project site for the duration of the contract. State of Florida water well construction rules, forms and reference documents can be found at <https://floridadep.gov/water/source-drinking-water/content/water-well-construction-rules-forms-and-reference-documents>. Rules are as prescribed by Chapter 62-531 F.A.C. and Chapter 62-532 F.A.C..

1.5.5 Installation Plan

Submit plan that includes, at a minimum: locating relief wells in the field; drilling, temporary and permanent casing; well assembly and jointing procedures; the temporary riser, implementing measurement datums, hanging the well in the borehole, method for placing filter pack, bentonite, and grout; well development; pump testing and sand testing, trimming of riser pipe; installation of check valves. Include the planned sequence of well installation with an explanation of crews and equipment proposed for work.

1.5.6 Shop Drawings

Submit shop drawings of each relief well components and installation devices detailing the following items:

- a. Well riser and screen assembly, including the length(s) and sequence of riser pipe and well screen used to construct each well, and;
 - (1) Location and type of couplings,
 - (2) The pipe diameter, well screen slot size(s) of each well screen and relief well pipe.
- b. Well centering device(s) used to center and align the assembled well in the borehole prior to placement of filter pack, and
- c. Filter pack tremie pipes and guide device(s) used to place the filter pack in the annular space between well screen and the wall of the borehole.

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Submit shop drawings for **surge block** device(s) used to mechanically surge the well in accordance with the paragraph entitled DEVELOPMENT. The Contracting Officer will not allow relief well installation to proceed until the required shop drawings related to that phase of the work have been submitted and approved.

PART 2 PRODUCTS

2.1 DRILLING FLUID

When drilling the boreholes for relief well, any drilling fluid additives used must be Liquid Polymer Emulsion (LPE) and treated with bleach and broken down prior to proceeding with the development process outlined below in the paragraph DEVELOPMENT. Bentonite or other expansive clay drilling fluid additives will not be allowed. Submit all **drilling fluid additives** to the Contracting Officer for approval prior to use. Dispose of drilling fluids in accordance with manufactures recommendations and local and State laws.

2.2 WELL SCREENS

Provide continuous slot, "V"-wire wrapped, stainless steel well screens fabricated by circumferentially wrapping a triangular-shaped wire around a circular array of internal bars. The wire configuration must produce inlet slots that are narrowest at the outer edge with the slot width increasing inwardly to minimize clogging. The nominal inside diameter of the well screen must be 10 inches. Slot openings must be uniform in width and be equally distributed around the entire circumference of the screen. The slot size and minimum open area for each type of screen is shown in the table below. The well screen must have a collapse strength of at least 170 psi and a recommended hanging weight of at least 14,500 lbs and equivalent to screens manufactured by Johnson Screens, St. Paul, MN 55164, telephone (612) 636-3900. Salvaged or previously used well screens, couplers, bottom plugs, or other well components of any kind will not be accepted. Provide manufacturer certification and documentation of **well screens** from the manufacturer that demonstrate the well screens and assembly meet all requirements requirements prior to delivery and assembly of any well.

TYPE	SLOT SIZE	MINIMUM OPENING AREA (square inch/foot)
No. 20 Slot	0.020 inch	48
No. 30 Slot	0.030 inch	68
No. 40 Slot	0.040 inch	85
No 50 Slot	0.050 inch	101

2.2.1 Stainless Steel

All pipes, rods, bars, and wires must be stainless steel Type 304 or 304L and conform to **ASTM A312/A312M** or **ASTM A555/A555M** as applicable.

2.2.2 Couplings and Joints

Fit all well screens with stainless steel coupling rings that are suitably

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beveled and finished to accommodate welded or threaded connections between lengths of riser or screen. Provide coupling rings from the same stainless steel alloy as the well screens and riser pipes. Ensure couplings conform in design to the couplings recommended by the manufacturer of the well screen. Factory weld coupling rings for lengths of well screen to ensure a secure and precise fit.

2.2.3 Bottom Plug for Well Screen

Fit each well assembly with a bottom plug consisting of a stainless steel plate. The diameter of the bottom plug must be equal to the outside diameter of the well screen. Factory weld the bottom plug to the bottom of the screen as described above in the subparagraph "Couplings and Joints".

2.2.4 Riser Pipe

Fit each well with a stainless steel riser pipe extending from the ground surface down to the top of the screen. Additionally, each well may require lengths of the riser or "blank" pipe to be inserted at various locations between lengths of well screen in accordance with the drawings. Riser and "blank" pipe must have a nominal inside diameter of 10 inches (NPS 10) and a nominal wall thickness corresponding to Schedule 10 stainless steel pipe (SCH 10S) or greater. Couplings between lengths of riser (including temporary riser) or "blank" pipe and between riser pipes and well screen must be consistent with the requirements set forth above in the subparagraph "Couplings and Joints".

2.3 FILTER PACK

Provide filter pack for the annular space between the borehole wall and the outside of the well screen meeting the following requirements:

- Comprised of naturally occurring grains and composed of hard, tough, durable particles free from adherent coating, and contain no soft, friable, thin, or elongated particles, or organic matter of any kind.
- From a source that can certify the site-specific manufacturing facilities are expressly qualified to produce and supply these materials for water treatment plants, in compliance with AWWA A100, AWWA B100 or NSF/ANSI 61 certified for quality and purity.
- Has an acid solubility less than 5 percent and specific gravity greater than 2.6.
- Has a coefficient of uniformity no greater than 2.5.
- Meets the following effective size requirements. Values are in millimeters:

Filter Pack	D10	D10min	D10max
20 Sand	0.57	0.53	0.6
30 Sand	0.87	0.79	0.9

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<u>Filter Pack</u>	<u>D10</u>	<u>D10min</u>	<u>D10max</u>
40 Sand	1.05	1.00	1.10
50 Sand	1.37	1.25	1.44

The individual grading curves within these limits must not exhibit abrupt changes in slope denoting skip grading, scalping of certain sizes or other irregularities that would be detrimental to the proper functioning of the filter pack. Materials must be uniformly distributed between sieves.

2.4 BENTONITE SEAL

Bentonite seal must consist of 1/2 inch or 1/4 inch bentonite chips.

2.5 NEAT CEMENT GROUT

Neat cement grout consists of 5.2 to 5.5 gallons of water per cubic foot or 94-pound sack of Portland cement, or a mixture of 6.0 gallons of water per sack of Portland cement with 3 to 7.5 pounds of bentonite not to exceed eight percent by weight.

2.6 GROUT

Use grouting and sealing materials that conform to [AWWA A100](#). Submit grout mix details and the location(s) the grout mix(es) will be used to the Contracting Officer prior to the use of any grout on the project.

2.7 CHECK VALVES

Fit each well with an aluminum check valve to prevent surface water and debris from entering the well riser. Fabricate the check valves in accordance with the details and dimensions indicated on the drawings and as specified herein. Make the check valves from one of the aluminum alloys 3005-H14, 6061-T4, or 6061-T6, and fabricate all the components of each individual check valve entirely from the same aluminum alloy. Welding of aluminum must conform to the applicable provisions of the [AWS D1.2/D1.2M](#).

The seat for the check valve must consist of a flexible neoprene rubber gasket sandwiched between two 1/4-inch aluminum plates, that are bolted together as indicated on the drawings. The neoprene rubber gasket must consist of solid (closed cell) neoprene rubber at least 1/4 inches in thickness and have a durometer hardness between 10A and 40A as measured on the "Shore A" scale in accordance with [ASTM D2240](#). Cellular neoprene or neoprene foam will not be accepted.

Weld the aluminum guide rods and aluminum lifting ring to the aluminum plates as indicated on the drawings. The aluminum guide rods must be dimensioned as indicated on the drawings.

Exercise extreme care to ensure that the assembled check valve forms a seal between the neoprene rubber gasket and the angular face ring and can slide freely in and out of the well discharge without binding or jamming. Also, take extreme care to ensure that the assembled check valve cannot shift to one side far enough to leave any portion of the well discharge open to the surface.

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2.8 ANGULAR FACE RING

Provide an angular face ring conforming to [ASTM A276/A276M](#), made from the same stainless steel alloy as the riser pipe and to the dimensions indicated on the drawings.

2.9 PERMANENT CASING

Provide permanent steel casing conforming to [ASTM A53/A53M](#) Grade B and has a wall thickness of 0.375 inches or greater. Submit documentation of the [permanent steel casing](#) to the Contracting Officer that identifies the source and ordering details of the permanent steel casing to be used for the project.

PART 3 EXECUTION

3.1 PRELIMINARY WELL LAYOUT

Prior to the commencement of any drilling operations, drive a wooden stake into the ground at each proposed relief well location to a horizontal accuracy of +/-0.3 feet. Label each stake with the corresponding well number and the existing natural ground elevation to a vertical accuracy of +/-0.1 feet. The Contracting Officer will review the proposed well locations and may adjust well locations in the field as needed. Under no circumstances will any drilling activity of any kind be permitted prior to the Contracting Officer's approval of the proposed well locations and confirm or re-confirm that each location is approved within 24 to 72 hours before mobilizing drilling equipment to the well. Maintain and/or replace these wooden stakes as needed to ensure that the approved well locations are clearly visible in the field until the relief wells are installed.

3.2 DRILLING

Before the drilling process begins, demonstrate that all materials, equipment, and personnel necessary to successfully install each well are present on-site and in good working order. All well drilling must be performed by a licensed well driller, licensed in the State of Florida. Provide the [well driller's license](#) to the Contracting Officer. Partial or total collapse of a borehole due to lack of material, unsuitable or defective equipment, inefficient operations, or negligence on the part of the Contractor will be considered cause for the Contracting Officer to order the Contractor to abandon the well and require the Contractor to re-drill the borehole at no additional cost to the Government. Locate and provide a suitable water source for use in drilling each well. Provision of water for drilling will be considered incidental to the drilling process and at no additional cost to the Government. Use only potable water for drilling, or water obtained from installed relief wells unless the Contracting Officer specifically states otherwise. Drill the wells in any sequence that begins at the limits of the wells on the south or west and in a manner that permits satisfactory groundwater management in accordance with subparagraph Groundwater. The Contracting Officer reserves the right to alter the sequence to achieve the objectives or for reservoir operation purposes. The drilling of new wells must not advance more than 5 wells beyond the satisfactory development and testing of the newly installed wells without approval from the Contracting Officer.

3.2.1 Groundwater

- a. The groundwater level in the construction limits is variable and is

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heavily influenced by the reservoir stage. Artesian pressures exist within the construction limits. Refer to Section 00 31 32 GEOTECHNICAL DATA REPORT for groundwater elevation readings for relevant piezometers and monitoring wells and important information on the subsurface and groundwater conditions. During the work, the reservoir will be operated up to elevation 36 feet NAVD88. Refer to Section 00 33 50 WEATHER AND WATER STAGE DATA for important information on reservoir operations. Anticipate that each relief well will flow freely at the top of riser (TOR) elevation defined in the drawings.

- b. Schedule and perform relief well installation in a manner that permits completed wells to provide groundwater pressure relief. Groundwater pressure relief is achieved by actively pumping a relief well, by gravity discharge following completion of the outfall, or both.
- c. Just before and periodically during the drilling process of each relief well, manually measure and record the groundwater elevation(s) of the nearest piezometer, piezometer nest, or installed relief well that will provide an indication of the groundwater level at the site of drilling.
- d. Perform drilling operations to maintain at least 3 feet of elevation differential between the level of the drilling fluid in the borehole and the natural groundwater, unless otherwise approved. Under no circumstances will the drilling of a well be allowed with an unstable borehole. Therefore, the Government will consider groundwater management, drilling methodology, and drill fluid management as a critical element to performing the work.

3.2.2 Temporary Piezometers

The existing piezometers within the construction limits are identified on the drawings and temporary piezometers are labeled "abandon". Protect all piezometers throughout the duration of construction and protect, then abandon temporary piezometers as follows. Protect the temporary piezometers throughout the duration of construction, or where the temporary piezometers co-exist with the installation of a relief well, abandon the temporary piezometers just prior to the drilling of the relief well. Following the installation of all relief wells, abandon the remaining temporary piezometers by removing as much of the casing pipe as possible but no less than 10 ft and allowing the borehole to collapse, then backfilling the remainder of the borehole with a suitable bentonite seal conforming the requirements specified in the paragraph BENTONITE SEAL above. Abandonment of temporary piezometers will be considered incidental to the installation of the relief wells.

3.2.3 Drilling Method

Drill all wells using the reverse rotary method or other industry typical method for water well drilling that is suitable for the conditions encountered. Use only approved drilling fluid additives as described above in the paragraph DRILLING FLUID. Other drilling methods (such as jetting or vibrating) that impart substantial disturbance of the aquifer, radical displacement of the formation, are unproven, not recognized by industry, or that may reduce the overall efficiency of the well are prohibited. Select drilling method to ensure acceptable placement of the well screen, riser pipe, filter pack, and satisfactory well construction. Dispose of cutting and other excess material resulting from the drilling process at the locations identified on the drawings, unless otherwise

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approved by the Contracting Officer. The diameter of the borehole must be sufficient to accommodate the diameter of the well screen and riser, along with the required minimum filter pack thickness indicated on the drawings.

3.2.4 Temporary and Permanent Casing

Protect the top of the borehole by a temporary steel casing. The inside diameter of the temporary casing must be sufficient to accommodate well construction in accordance with this section and as indicated on the drawings. The wall thickness of the temporary casing must be sufficient to ensure that no significant deformations occur anywhere along its length throughout the well installation process. At all times and until removed, anchor the temporary casing securely to the drill rig or ground surface. Install all casing in a borehole that is equal to or less than the inside diameter of the casing. The borehole above the filter pack must remain cased until the well has been sealed and grouted. After drilling, if groundwater may rise above top of temporary casing, additional pipe must be added to contain groundwater within the casing, but below the well riser. After the well is completed, a portion of the temporary casing must remain as permanent casing. Provide permanent casing in accordance with paragraph PERMANENT CASING. The top of the permanent casing is to be finished to within one foot of the base of the manhole and must not be modified until the manhole is ready to be placed. The bottom of the permanent casing must be installed to at least elevation 12, but no lower than elevation 10.

3.2.5 Obstructions

The equipment used to drill the well borehole must be capable of drilling through materials identified at the project site and described in Section 00 31 32 GEOTECHNICAL DATA REPORT. This includes equipment having the capability of drilling through and removing rock fragments, cobbles, fossils and/or shells that can exceed 6 inches in width (in-place). The presence of obstructions such as these will not be considered sufficient grounds for abandoning a borehole. If obstructions are encountered during drilling which, in the opinion of the Contracting Officer, render it impracticable to complete drilling of the borehole, the Contracting Officer may adjust the depth of the well accordingly. Alternatively, the Contracting Officer may direct the Contractor to abandon and backfill the borehole, plug the hole in accordance with the paragraph below entitled PLUGGING ABANDONED HOLES and drill another borehole within a 25 feet radius.

3.2.6 Drill Fluid Pit Systems

Excavation for drill fluid pit systems will not be permitted. If used, drilling fluid pits must be portable. Maintain a minimum operational capacity equal to three times the volume of the borehole to be drilled and incorporate desanders or baffles to separate fine material from the drilling fluid.

3.3 INSTALLATION OF RISER PIPE AND SCREEN

Maintain drilling fluid, any additive(s) and any temporary casing to provide a stable and open borehole while performing the work.

3.3.1 Storage

Once delivered into the Contractor's custody, store all well components

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(including but not limited to riser pipes, screens, couplers, bottom plugs, well guards, check valves, and other fittings) to ensure that they are not damaged, degraded, clogged, or otherwise impaired until they are properly installed. When the Contracting officer makes a request to inspect well components, make all well components available for visual inspection within 24 hours of the request.

3.3.2 Assembly

Do not splice shorter segments of riser pipe or screen less than 15 ft in length together to form longer lengths of riser pipe or screen unless a variance is requested and approved by the Contracting Officer. This does not limit the use of couplings where the relief well profile calls for a change in well screen or riser pipe. Precisely mark and score the boundary between the temporary riser and TOR shown in the plans. Measure and verify the correct materials are assembled and the assembled lengths match the required lengths of riser and screen indicated in the contract drawings to a tolerance no more than 0.05 foot.

3.3.3 Joints

Joints between lengths of riser pipe may be threaded or be either single or double bevel welded and must have strengths equal to or greater than the pipes being joined. Fully seat threaded joints. Weld metal for welded joints must have a corrosion resistance equal to or greater than that of the pipe or screen used. Design and construct joints to support the weight of the screen and/or pipe as it is lowered into the hole. Welding must be performed by a certified welder using appropriate welding rods. Exercise extreme care to ensure that the well assembly (i.e. riser pipes and screens) retains its circular cross-section and straight alignment both during and after the welding process. Also exercise extreme care to ensure that slag or other weld metal is not deposited inside the circumference of the pipe that would obstruct the path of a pump, surge block, or other likely equipment lowered into the well.

3.3.4 Installation

Install each properly assembled well straight and plumb within the borehole. Handle all well components with care and keep free of any structural damage, degradation, clogging, or contamination throughout the construction process. If, in the opinion of the Contracting Officer, any component of the well (including but not limited to riser pipes, screens, couplers, bottom plugs, well guards, check valves, and other fittings) is damaged, degraded, clogged, or contaminated during any stage of the construction process, the Contracting Officer may require the Contractor to repair, clean, or replace that component at no additional cost to the Government. Damage identified after the well has been installed in the borehole will not be exempt from this requirement. Take extreme care to prevent any mud, sand, organic materials such as grass, and other contaminants from entering the well borehole during the installation process. Take extreme care to minimize deformation of the well screen during handling and installation. Take extreme care to avoid impacts between the well components and the sides of the borehole during the installation process. Suspend the well screen in the hole and never allow it to rest on the bottom

3.3.5 Centering Device

Employ an approved device(s) to center and align the well within the

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borehole throughout the installation process. Build each centering device so that it either attaches to the bottom of the tremie pipes used to place the filter pack or allows the tremie pipes to pass freely up and down through the centering device. An acceptable centering device must have a difference no more than 1-1/2 inches between the diameter of the borehole and the outside diameter of the centering device. The centering device is not required to be continuous around the entire circumference of the well, but it is required to ensure proper clearance between the well assembly and the borehole wall in both perpendicular directions (e.g. north-south and east-west). Design the centering device to minimize disturbance of the borehole walls during use, and its leading and trailing edges be suitably rounded or beveled to minimize gouging the borehole wall as the device is raised and lowered.

3.3.6 Securing the Well

Immediately after the assembled well has been successfully inserted, centered, and hung in the borehole, fasten the top of the well securely both horizontally and vertically. Secure the top of riser (TOR) to within 0.05 foot of the elevation provided in the contract drawings and until the entirety of the filter pack has been tremied into place. Provide means to check and verify the TOR using the temporary riser (or other approved means) to achieve this requirement.

3.3.7 Depth Test

Immediately after securing the well, perform a depth test using an ultrasonic sounding device or weighted tape measure. If the depth measurement indicates excessive deformation or improper assembly of the well, the Contracting Officer may require the Contractor to perform additional testing to ensure that the well was properly assembled. If the well is found to be excessively elongated, the Contracting Officer may require the Contractor to replace the well at no additional cost to the Government. If the well is found to be improperly assembled, the Contracting Officer may require the Contractor to replace the well at no additional cost to the Government.

3.3.8 Plumbness and Alignment Test

Each well must be sufficiently straight, such that a cylinder 10 feet in length and with an outside diameter 1/2 inch smaller than the inside diameter of the well may be lowered for the full depth of the well and withdrawn without binding against the sides of the well. Furnish the dummy cylinder and perform the alignment check; however, this will not relieve the Contractor of the responsibility of maintaining adequate clearance for installation of the surging and pumping equipment required for testing and pumping the wells. Perform at least one alignment check on each well after placement of the filter pack. Additional tests may be made during the performance of the work at the option of the Contractor.

3.4 PLACING FILTER PACK

Maintain drilling fluid additive(s) and provide a stable and open borehole while performing the work herein.

3.4.1 General

Stockpile or store the approved filter pack bags to be used for each well adjacent to that well location in numbered, traceable lots and protect it

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from contamination or degradation. Filter pack that has not been tested and approved will not be allowed. Do not use filter pack that has come in direct contact with the ground. Once each well assembly has been hung in the borehole and the depth, alignment, and plumb tests (if required) have been completed, tremie the filter pack stockpiled at that well into the annular space between the well screen and the borehole wall.

3.4.2 Tremie Pipe

Tremie pipes used to place filter pack must consist of 10-foot lengths of standard 1-1/2-inch inside diameter (I.D.) pipe or larger positioned on opposite sides of the well. Each pipe must have slotted openings 1/16 inch wide spaced equally around the circumference of the pipe to help prevent binding of the filter pack as it slides down the tremie pipes. The tremie pipes must be free of any dents, holes (other than the slots described above), flat spots, damaged threads, or burrs over their entire length. Feed tremie pipes from a hopper positioned above the well and balanced so that filter pack can slide freely into both tremie pipes. Design connections between the tremie pipes and the hopper for quick connection and disconnection so that adding and removing tremie pipes can be accomplished with the least possible delay.

3.4.3 Tremie Guide

Hold the tremie pipes in position on opposite sides of the well assembly by means of a tremie guide. The tremie guide must consist of a metal ring(s) with an inside diameter that will permit it to slide freely up and down full length of the well assembly. Fit the tremie guide with a pair of collars capable of mating with the tremie pipes. Maintain the bottom of the tremie pipes through which the filter pack will exit into the annular space at a distance of 10 feet from the tremie guide throughout the process. This tremie guide may be used as a centering device to meet the requirements set forth above in the subparagraph "Centering Device" above provided that the device is built in such a way that it meets all the requirements for both the centering device and the tremie guide.

3.4.4 Filter Pack Placement

- a. Throughout the entire process of placing filter pack, shield or cover the top of the borehole and well riser to prevent filter pack or other debris from falling into the borehole except through the tremie pipes. Be prepared to fill the hopper quickly and efficiently in any position from the ground surface up to the maximum height at which the hopper will be set. Mix the filter pack as necessary to minimize segregation prior to feeding it into the hopper. The Contractor may also elect to supply water to the hopper to help ensure a continuous, smooth flow of filter pack from the hopper into the annular space.
- b. At the commencement of the operation, lower the tremie pipes down until they come to rest on the bottom of the borehole. Then slowly raise the tremie pipes in regular increments, allowing the filter pack to slide down and fill the annular space between the well and the borehole wall.
- c. At all times during the placing of the filter pack, keep the hopper and tremie pipes filled with filter pack to 5 feet above the top of the well.
- d. Do not allow the bottom of the tremie pipes to be more than 2 feet

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above the top of the filter pack already placed in the borehole.

- e. Do not raise, lower, or support the tremie assembly by only one pipe. If the borehole is stabilized with temporary casing, ensure that the borehole remains stabilized and casing is extracted incrementally as the filter pack is tremied into position.
- f. Do not allow the bottom of the borehole casing to be less than 1 foot below the top of the filter pack and extract the casing in 2-foot increments behind the filter pack placement. Continue the alternating placement of filter pack and extraction of tremie pipes and casing without interruption. Continuously execute the alternating placement from the bottom of the borehole until the filter pack has reached or exceeded the level indicated on the drawings.
- g. Provide a reliable and repeatable means of measuring the depth from a suitable measurement point to the top of the filter pack during placement. Measure and record the depths and ensure they are in accordance with the plans and tolerances prescribed by this section.
- h. Once the filter pack has been placed, perform the test as prescribed by subparagraph "Plumbness and Alignment Test" above.

3.5 DEVELOPMENT

- a. Once the filter pack placement requirements have been met and any drilling fluid additives broken down, develop each relief well to produce a stable well with maximum efficiency. Development of the newly installed wells must keep pace with the installation of new wells as defined in paragraph DRILLING above.
- b. Start the development process no more than 5 days following the installation of the filter pack and perform steadily and continuously until the development process for that well is complete.
- c. Maintain access to the well throughout the development process.
- d. Install, maintain, adjust, or enlarge drainage as needed throughout the duration of the contract to ensure that no unwanted discharge of drilling fluids, development water, stormwater, sediment, or debris occurs anywhere outside the project Right-of-Way (ROW).
- e. If, at any time during the development process, the Contracting Officer judges that the well may be in danger of being damaged, clogged, or improperly developed, the Contracting Officer may order the development process to be extended, altered, or stopped altogether.
- f. The Contracting Officer may require a change in method if the Contractor's development method fails to accomplish the desired results.
- g. The Contracting Officer may order that wells that continue to produce excessive amounts of fines after development be abandoned, plugged, and backfilled, and may require the Contractor to construct new wells nearby.
- h. Provide a bubbler tube or other approved means for accurately determining the water level in the well under all conditions.

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- i. While surging or pumping, monitor the top elevation of the filter pack and maintain it such that the filter pack does not drop below the bottom of temporary casing used to support the borehole above the filter pack.
- j. After each surging or pumping cycle, add the designated filter pack to the annular space around the well to the required elevation shown on the contract drawings.

3.5.1 Mechanical Surging

To remove any sediment or other fines that may have accumulated on the borehole wall or within the filter pack, subject each well to a period of mechanical surging.

3.5.1.1 Surge Block

- a. Furnish a surge block that consists of at least 2 sets of circular, neoprene rubber disks rigidly affixed to a central shaft comprised of a drill stem or other heavy duty pipe of sufficient mass to cause the surge block to free fall rapidly down the well under its own weight.
- b. Use 1 inch thick neoprene rubber disks that have a durometer hardness between 40A and 70A as measured on the "Shore A" scale in accordance with [ASTM D2240](#). Cellular neoprene or neoprene foam will not be accepted.
- c. Make each set of circular, neoprene rubber disks from a 9-inch diameter neoprene rubber disk sandwiched between two 8-inch diameter neoprene rubber disks. Support each set of neoprene rubber disks at each end using rigid metal washers 2-1/2 to 3-1/2 inches in diameter.
- d. Space the two sets of neoprene rubber disks 4 feet apart on the central shaft.
- e. Adjust the diameter of the rigid metal washers, the weight of the central shaft, or the rate of travel of the surge block as directed by the Contracting Officer to achieve a reliable, strong surging action without damaging the well screen or filter pack.
- f. Replace the neoprene rubber disks as soon as they show evidence of wear or degradation.
- g. Furnish any equipment necessary to raise and lower the surge block at a rate between 1-1/2 and 3 feet per second (ft/s) on both the up and down directions over the full length of the well.

3.5.1.2 Mechanical Surging Procedure

- a. Prior to inserting the surge block into the well, measure and record the depth of the well to the nearest 0.1 foot. Compare this measurement to the design depth of the well provided on the drawings.
- b. Remove any measurable sediment or debris found on the bottom of the well by air lifting prior to the start of the surging process.
- c. Then subject each well to at least 3 cycles of surging.
- d. Each cycle must consist of placing the surge block at the top of the

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well screen, allowing the surge block to free fall for 20 feet, raising the surge block at a rate between 1-1/2 and 3 feet per second (ft/s), and repeating this action.

- e. Then lower the surge block to 10 feet below the top of the well screen and the surging process resumed.
- f. In this way, surge the entire well, beginning at the top and progressing toward the bottom to prevent the surge block from becoming sand locked, over a period of at least 60 minutes.
- g. This 60-minute procedure covering the entire well will constitute 1 cycle of surging.
- h. Allow at least 5 minutes between cycles of surging to allow sand and other sediment to settle to the bottom.
- i. After 5 minutes, measure and record the depth of the well to the nearest 0.1 foot, remove any sediment that may have accumulated by injecting compressed air at the bottom of the well at a rate sufficient to produce a discharge of 100 gpm for at least 30 minutes, and then repeat the surging procedure, beginning at the top of the well screen.
- j. Exercise care to prevent the surge block from impacting the bottom of the well during the down-stroke when surging near the bottom of the well.
- k. If surging generates material that blocks 10% or more of the screen, cease surging and remove material pulled into the well before continuing the surging procedure.

3.5.2 Pump Surging

- a. After each well has been mechanically surged, subject it to several periods of intense pumping.
- b. Measure the depth from the top of the well down to the static groundwater level, lower a centrifugal or turbine type pump to within 10 feet of the bottom of the well, and pump the well at a flow rate sufficient to produce at least 5-10 feet of drawdown for a period of 5 minutes.
- c. After 5 minutes, shut off the pump and use a valve to prevent any backflow through the pump from re-entering the well.
- d. When the water level in the well has rebounded back to its original static level, measure and record the depth of the well to the nearest 0.1 foot, air lift out any sediment that may have accumulated, and repeat the pumping process.
- e. This process of pumping for 5 minutes, allowing the water level in the well to rebound, and measuring the depth of the well will constitute 1 cycle of pump surging. Repeat this cycle of pump surging continuously for a period of at least 1 hour, at the end of which, air lift any remaining sediment out of the well.

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3.5.3 Well Development Data

Measure and make observations throughout the development process for each well and record the results on the "Well Development Data" form provided in Attachment A. At a minimum, record the following items for each well:

- a. Start and end dates of well development,
- b. Start and end times of each cycle of mechanical surging and pump surging,
- c. Datum for depth measurements and its associated elevation,
- d. Depth to the static groundwater level at the beginning of well development,
- e. Depth to the top of the filter pack before and after each cycle of mechanical surging or intermittent pumping,
- f. Depth to the bottom of the well before and after each cycle of mechanical surging or intermittent pumping.
- g. Amount of filter pack added and when it was added during the development process.
- h. Confirmation that the filter pack is at the required top elevation as shown in the plans at the completion of development.
- i. Any other information necessary to demonstrate compliance.
- j. Photos of the water samples at the start, middle, and end of the development process.

3.6 PUMP AND SAND TESTING

Once each well has been developed, but before acceptance, perform a pump test and sand test on each well. Perform a constant rate pump test on all wells. During the course of the pump test, also perform a sand test to measure the quantity of sand in the pump discharge. For a pump test to be accepted, the water level measured in the well must remain above the top of the well screen throughout the entire duration of the test, unless specifically authorized by the Contracting Officer.

3.6.1 Equipment

Furnish the following equipment to perform each pump test.

3.6.1.1 Pump

Provide a submersible turbine or centrifugal type pump capable of pumping at least 150 gpm against a total dynamic head of approximately 50 feet. Furnish the motor and power source (e.g. electrical, diesel, gasoline, etc) to run the pump as required.

3.6.1.2 Flow Meter

Furnish and install a calibrated flow meter capable of accurately measuring the discharge rate from the well to the nearest 1 gpm and capable of totalizing the accumulated quantity of discharge over time.

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Documentation of the flow meter calibration must be readily available to be submitted at the Contracting Officer's request.

3.6.1.3 Flow Control Valve

Furnish and install a flow control valve of standard design capable of regulating the discharge rate during the step drawdown pump tests. However, the Contractor may elect to use a variable speed pump to achieve the required flow rates. If so, the flow control valve may not be required.

3.6.1.4 Water Level Measurements

Furnish a device capable of accurately measuring the water level in the well to the nearest 0.01 foot during the pump test and digitally records the reading at a frequency of at least one every minute. A weighted tape or water level indicator, or bubbler tube will not be considered appropriate for use during the pump tests but are necessary for verifying and supplementing the digital record.

3.6.1.5 Sand Infiltration Measurements

Furnish and install a Rossum centrifugal sand tester or an approved equal capable of measuring the sand content of the pump effluent during the pump test in accordance with AWWA A100, Appendix E, Section E.4.6 "Measurement of Sand Content". Set up and operate the sand tester in accordance with the manufacturer's instructions. Equip the pump discharge line with the necessary fittings to properly conduct the test. The Contractor's employee must be experienced in conducting the sand infiltration test including placing the sampler at the proper location along the discharge line, calibrating the proper flow into the meter, and experienced in reading the amount of sand in the sampler.

3.6.2 Pump Testing Procedure

- a. Perform the pump tests and the associated sand tests under the supervision of the Contracting Officer.
- b. Test each well by pumping continuously for a period of at least 1 hour at flow rate of 100 gpm.
- c. Employ a flow control valve to dial in the required flow rate to within +/-10 gpm.
- d. Adjust the flow control valve as needed throughout the test to ensure a constant flow rate for the entire duration of each step.
- e. Measure the water level in the well to the nearest 0.01 foot and recorded at each of the following time steps (in minutes): 0, 1, 2, 3, 4, 5, 10, 15, 30, 45, and every 15 minutes thereafter, until water levels in the pumped well have stabilized, as demonstrated by a straight-line trend on a plot of drawdown versus the logarithm of time since pumping began, or as directed by the Contracting Officer. Assume the minimum continuous pumping time for each pumping rate is 60 minutes and the maximum time is 180 minutes.
- f. If the well is identified for detailed pump testing, then perform the procedure defined in subparagraph "Detailed Pump Testing Procedure".

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3.6.3 Sand Testing Procedure

Measure the sanding rate of the pumped well during the last 30 minutes of each "step".

If the sand infiltration rate measured during the final "step" is less than 5 parts per million (ppm) over a 15 minute period, the well will be considered acceptable with respect to sanding.

If sand is visible in the sampler, but it does not reach up to the first readable unit of measure, then do not try to estimate the quantity of sand in the bottom of the sampler. In this case, record the reading as "Trace".

If the sand infiltration rate measured during the final "step" exceeds 5 ppm over a 15 minute period, then subject the well to additional rounds of pump surging as described above in the paragraph entitled DEVELOPMENT until the sanding rate has been reduced to an acceptable level.

If at the end of 8 hours of additional pump surging, the sanding rate is still unacceptable, abandon the well. However, the Contractor may elect to continue pump surging or take other remedial measures at his own expense in to attempt to reach an acceptable sanding rate. If a sand test is performed in the presence of the Contracting Officer and the sanding rate is found to be less than 5 ppm, then the well will be considered acceptable.

3.6.4 Detailed Pump Testing Procedure

Perform a "step drawdown" pump test on each of the detailed pump test wells identified on the plans. This procedure is in addition to the pump testing and sand testing procedures defined in this section, except where those requirements are modified by this paragraph. Do not perform this procedure until all relief wells and outfalls within 500 feet of the pump well are recognized as completed by the Government and no dewatering is occurring within this radius. Pump continuously for a period of 1 hour at each of the following flow rates (i.e. "steps"): 33 gpm, 66 gpm, and 100 gpm. Provide a riser or packer to prevent flow from any relief well within 500 ft of the pump well. Provide all information required of subparagraph "Pump and Sand Testing Data."

3.6.5 Pump and Sand Testing Data

Record all pump test and sand test results and testing measurements on forms similar to the "Relief Well Pump Test" and "Sand Infiltration Test" forms provided in Attachment A and submit these records on paper and in electronic format (.PDF and .XLSX) in an orderly, organized way for each relief well to the Contracting Officer. At a minimum, submitted items must include the following:

- a. The well ID station, northing, easting, and elevation of the well being tested,
- b. Date and time each pumping test began,
- c. Reservoir and canal water level at the start of the test,
- d. The elapsed time at which each water level reading was recorded,
- e. The depth to water (to the nearest 0.01 foot) in the well prior to

starting the pump test,

- f. The depth to water (to the nearest 0.01 foot) in the well at each required time increment,
- g. The pump discharge rate (to the nearest 1 gpm) at each required time increment,
- h. The name, station, northing, easting, elevation and depth to water (to the nearest 0.01 ft) of any piezometers along the line of wells that are within a 500 ft. radius and at least two relief wells within 50 to 200 feet (divided equally to either side of the pumped well). This requirement only applies to the "Detailed Pump Testing Procedure,"
- i. The quantity of sand measured in the Rossum sand tester at the end of the time interval for the designated flow rate(s),
- j. The depth to sand in the well immediately before and after the "step drawdown" test,
- k. Notes regarding any sanding problems or additional pump surging required to produce an acceptable well.
- l. Details of any pumping or dewatering occurring within a 500-foot radius.
- m. Any other information deemed necessary to evaluate the tests or for the Government to perform independent evaluation of the test results.

3.7 WELL INSTALLATION DIAGRAM

Prepare and maintain a well installation diagram for each well prior to the start of drilling through the pump test and sand test. Immediately following the pump test and sand test and prior to the Video Inspection, submit the [well installation diagram](#). This well installation diagram must contain all depth (referenced to the temporary measurement datum), elevation and dimensional information associated with its installation, included but not limited to:

- a. Station/offset, northing, easting,
- b. Datum for depth measurements and associated elevation,
- c. Drilling method,
- d. The assembled length of well string at time of installation,
- e. Any use of temporary casing, including its extents and diameter,
- f. Drillers log and notes, including conditions encountered and any issues encountered during drilling,
- g. Total depth of drill hole,
- h. Bottom and top of any filter pack and pipe type change,
- i. Quantity of filter pack(s) used and reference to its lot and approved submittal(s),

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- j. TOR elevation, as measured at the scored mark at the boundary of the temporary riser and TOR,
- k. Borehole diameter.

3.8 VIDEO INSPECTION

Perform a downhole video inspection of each well. Record the inspection as the camera descends. Descend the downhole camera in the center of the well, at a rate that provides a clear video with minimal shaking, no faster than 2 feet/minute, directly facing the sidewall while rotated at a speed that permits clear inspection of the well's interior circumference, and at a slow steady speed. The video must be high-definition digital video (at least 1280x720 resolution at a minimum of 30 frames per second). The device must be fitted with dual cameras (downhole and side view) and be able to continuously pan. The camera must also be self-focusing with the ability to adjust setting such as focus, orientation, exposure, and lighting intensity to provide a plainly visible video. It must also be equipped with a footage counter to measure depth with a measurement error of less than 1%. The well name, date, and camera depth (at a minimum) must be visible on the video and during the inspection. Record depths of features such as screen type or blank sections, foreign objects, deformations, or fractures. If the well screen is not plainly visible due to biofouling or other sediment, brush and pump the well and reshoot the video. Provide the locations of all features and make side by side comparison of the well screen depicted on the installation diagram and the depths and elevations obtained from the video inspection. Make note and explanation of any discrepancies. Submit all documentation for the [video inspection](#) to the Contracting Officer for approval.

3.9 CHLORINE TREATMENT

Following the video inspection, kill any unwanted bacteria that may have been introduced to the well during construction or development, dissolve a calcium hypochlorite compound (for example, HTH) with a minimum 70 percent available chlorine in a volume of water equal to the volume of the well and the resulting solution poured into the well. The concentration of calcium hypochlorite must be 1 pound per 100 gallons of solution. Agitate the chemicals in the well by slowly and continuously injecting compressed air at the bottom of the well for a period of 2 hours followed by a period of 6 hours where the water in the well is undisturbed to allow sediment and debris to sink to the bottom. Repeat this cycle of injecting calcium hypochlorite solution, agitating for 2 hours, and resting for 6 hours continuously for a period of 24 hours. At the end of the 24 hour period, carefully remove any material in the bottom of the well by air lifting with an eductor pipe for 20 minutes and direct discharge to an appropriate location. Other methods of agitating the chemicals besides compressed air or removing sediment may be used if approved by the Contracting Officer. Do not discharge water or chemicals from the well at any time during the entire 24-hour chemical treatment process.

3.10 BACKFILLING

Once the pump test and sand test has been completed, all prior submittals provided and approved for the well, and the filter pack top elevation verified, place 2 feet of bentonite on top of the filter pack. Hydrate bentonite for a minimum of 30 minutes. Use bentonite conforming to the requirements of paragraph BENTONITE SEAL. Take extreme care to prevent

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disturbance to the filter pack and creation of voids in the bentonite seal. Ensure that the entirety of the annular space is filled with competent, impervious material. Once the bentonite is in-place, place grout on top of the bentonite seal. Use material conforming to the requirements of paragraph GROUT. Take extreme care to prevent bentonite, grout, or displaced groundwater from entering the well through the top of the well riser pipe.

3.10.1 Grout Placement

Place grout under pressure from the bottom upward in a continuous operation to ensure a complete seal of the annular space in accordance with Appendix C of [AWWA A100](#). Maintain the well riser in the center of the borehole. When mixed at the job site, use within 45 minutes of mixing. Grout mixed at a commercial mixing plant and transported to the job site in trucks must be used in the work within 1-1/2 hours after mixing. Retempering of grout will not be allowed.

3.11 INSTALLATION OF ANGULAR FACE RING

Cut off the top of the riser pipe 1/8 inch below the well discharge elevation (i.e. TOR) provided in the contract drawings. Exercise extreme care to ensure that the cut is square, smooth, and even around the entire circumference of the riser pipe. De-bur, grind, file, or brush the cut as necessary to ensure a clean, smooth rim around the top of the riser pipe. Weld an angular face ring to the top of the riser pipe as shown on the contract drawings to provide a continuous, level rim set 1/8-inch above the cut end of the riser pipe. Use the 1/8-inch difference between the top face of the ring and the cut end of the riser pipe to accommodate the weld. The weld must be continuous around the entire inside circumference of the angular face ring and be ground smooth once it has been completed. The angular face ring itself must consist of a continuous, stainless steel ring made from the same alloy as the riser pipe and securely fixed with its horizontal surface set at the well discharge elevation provided in the drawings. Both the vertical and horizontal legs of the ring must be at least 1 inch long, and both legs must be at least 1/8 inch thick as indicated on the drawings. Exercise extreme care to ensure that the top surface of the angular face ring is perfectly level and within 0.010 foot of the elevation indicated in the drawings, and completely free of any imperfections, welding slag, cuts, gouges, or other imperfections that might prevent the check valve from seating properly around the entire circumference of the ring.

3.12 INSTALLATION OF CHECK VALVE

Protect each well discharge by a check valve as indicated on the drawings. Construct the check valve as described above in the subparagraph "Check Valves" and insert into the well discharge immediately after the borehole has been sealed as described above in the paragraph BACKFILLING. If check valves are not available at this time, the Contracting Officer may approve the use of a temporary cover for the well to be used until the check valves can be installed. Never leave the well riser uncovered for any reason. Exercise extreme care to ensure that no trash, debris, dirt, vegetation, stormwater, or other contaminants that might clog or impair the well are allowed to enter the well at any time. Exercise extreme care to ensure that the check valve remains intact and free from any spattered concrete, dirt, debris, or other obstructions at all times. Each check valve will be inspected by the Contracting Officer to ensure it is in place, free from damage or defect, and functioning as

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intended. Repair or replace any check valve deemed unacceptable by the Contracting Officer at no additional cost to the Government.

3.13 PLUGGING OF ABANDONED HOLES

Fill wells designated by the drawings or the Contracting Officer to be abandoned with a grout conforming to the requirements of paragraph GROUT. Place grout under pressure from the bottom upward in a continuous operation to ensure a complete seal of the annular space in accordance with Appendix C of **AWWA A100**. Note that for abandoned wells, the use of bentonite pellets or chips will not be allowed. Prior to grouting, recover any well screen, riser pipe, or other well components that the Contracting Officer considers salvageable if possible. Cut off the riser pipe at least 3 feet below the ground surface and grout the remaining well and borehole up to within 3 feet of the ground surface using a tremie pipe to be approved by the Contracting Officer. Position the bottom of the tremie pipe 3 feet from the bottom of the well while the grout is being injected. Inject the grout at a maximum pressure of 0.5 pounds per square inch (psi) per linear foot of abandoned well. Once the abandoned well has been fully grouted, backfill the remainder of the hole with natural soil and thoroughly compact to prevent a rut or hole from forming at the abandoned well site.

3.14 EMERGENCY MEASURES

Never allow drilling operations to exert borehole pressures that could cause borehole blowout in the perimeter canal. Immediately report any rapid and continual drill fluid losses to the Government, implement measures to stop the rapid and continuous loss. If the losses are the result of or could be the result of borehole pressures causing blowout in the perimeter canal, immediately fill the borehole with filter pack in the Government's presence. If the Government suspects this to be the case but the borehole has not been filled, the Contracting Officer reserves the right to direct the Contractor to fill the borehole with filter pack at no additional cost. Following the immediate repair with filter pack, the Government reserves the right to direct reasonable means to explore and repair the area the blowout occurred. Whether directed or not, the Contractor is responsible for protecting the work from foreseeable events.

3.15 PROJECT EQUIPMENT

Furnish and fabricate the following equipment, in new or like new condition, and provide the equipment and associated materials to the Contracting Officer. Provide equipment that is the same type and brand as the equipment used to perform the work required under paragraph DEVELOPMENT and paragraph PUMP AND SAND TESTING (except for the air compressor used for air lifting). Fabricate and build the assembly of equipment that is functional, serviceable, and suitable for use in a way that the Contractor would use to test, clean, and maintain the operation of the relief wells. Such equipment and assembly also includes:

- a. Surge block, in accordance with sub-paragraph Surge Block and 20 sets of neoprene rubber discs.
- b. Electric crane/hoist sufficient for raising and lowering the surge block and pump.
- c. Tandem axle trailer. Each axle rated for 7,500 lb. with electric or hydraulic assist trailer breaks.

- d. Storage tank for chlorine treatment with 5 to 10 gallon per minute transfer pump and discharge hose.
- e. 6 inflatable packers capable of plugging the riser pipe and manhole outfall pipe with 10 feet of head.
- f. Electric or gas air compressor that has a minimum 20-gallon capacity.
- g. Heavy duty powder coated aluminum or steel tool storage box that is weather resistant, lockable and designed for outdoors.
- l. Other accessories necessary to make the trailer road worthy and equipment fully functional for its intended use.

After delivery to the Contracting Officer, demonstrate its function and capabilities to the Government. The demonstration must be as the work was performed on other wells and in the presence of the Government. Unless otherwise approved, the demonstration is not to be performed until all relief wells are constructed. The Government will accept the project equipment after successful demonstration and delivery of all the necessary project equipment covered under this paragraph.

3.16 REGULATORY

Register each well with the state/county and provide all required documentation to the state, as required by the state/county.

-- End of Section --

Project:			Well ID:			Date:		
Technician/Company:			Well Station:					
Target Pump Rate:								
Start Time:								
Test Number	Time (min) (start/stop)	Flowmeter Reading (Flow) (start/stop)	Pumping Rate (GPM)	Elapsed Time (min)	Rossum Outlet Valve (gmp)	Sand Tube Reading (ml)	Incremental (ml)	Sand Content (ppm)
Notes: -Rossum outlet valve should be calibrated to 0.5 GPM. -Rossum Reading (ml)/ Length of Test (min) = Sanding Rate (ml/min). Then, Sanding Rate (ml/min) x 528 = Sanding Rate (ppm)								
If sanding rate exceeds specifications, describe measures taken between tests:								
Additional remarks:								
Testing complys with specifications and manufacture instructions:								
					_____ Technician			

Relief Well Pump Test

Project:	Well ID:	Date:
Technician/Company:	Well Station:	
Target Pump Rate:	Instrumentation Used:	
Start Time:		

Static/Initial Water Data Information Prior to Test					
Location	ID	Time of Recorded Level	Elevation of Measurement Datum	Depth to Water Level (feet)	Water Level Elevation
Test Well					
Reservoir			N/A	N/A	
Perimeter Canal			N/A	N/A	

Specific Capacity			
Pump On Time	Pump Off Time	Duration of Pumping	Target Pump Rate

Target Time Since (min)	Actual Time Since (min)	Pumping Rate (GPM)	Measurement (feet)	☐ Depth to Water ☐ Depth of Water	Elapsed Time (min)	Drawdown (feet)
1						
2						
3						
4						
5						
10						
15						
30						
45						
60						
1		Off				
2		Off				
3		Off				
4		Off				
5		Off				
10		Off				
15		Off				
30		Off				
45		Off				
60		Off				

Time of Last Measurement: _____

Additional remarks:

Notes:

- Attach automated data records and plot of drawdown versus logarithm of time since pumping began.
- Attach sand infittation test results.

Testing complys with specifications and manufacture instructions:

_____ Technician

[illegible]